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NKT cells co-expressing a GD2-specific chimeric antigen receptor and IL-15 show enhanced *in vivo* persistence and antitumor activity against neuroblastoma.

Xin Xu¹, Wei Huang¹, Andras Heczey^{1,2,3}, Daofeng Liu¹, Linjie Guo¹, Michael Wood¹, Jingling Jin¹, Amy N. Courtney¹, Bin Liu¹, Erica J. Di Pierro¹, John Hicks¹, Gabriel A. Barragan¹, Ho Ngai¹, Yuhui Chen⁴, Barbara Savoldo⁴, Gianpietro Dotti⁴, Leonid S. Metelitsa^{1,2,3}

¹Texas Children's Cancer Center, Department of Pediatrics, Baylor College of Medicine, Houston, TX, 77030

²Department of Pathology and Immunology, Baylor College of Medicine, Houston, TX, 77030

³Center for Cell and Gene Therapy, Baylor College of Medicine, Houston, TX, 77030

⁴Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599

Abstract

Purpose: V α 24-invariant natural killer T cells (NKTs) are attractive carriers for chimeric antigen receptors (CARs) due to their inherent antitumor properties and preferential localization to tumor sites. However, limited persistence of CAR-NKTs in tumor-bearing mice is associated with tumor recurrence. Here, we evaluated whether co-expression of the NKT homeostatic cytokine IL-15 with a CAR enhances the *in vivo* persistence and therapeutic efficacy of CAR-NKTs.

Experimental design: Human primary NKTs were *ex vivo* expanded and transduced with CAR constructs containing an optimized GD2-specific single-chain variable fragment and either the CD28 or 4–1BB costimulatory endodomain, each with or without IL-15 (GD2.CAR or GD2.CAR.15). Constructs that mediated robust CAR-NKT cell expansion were selected for further functional evaluation *in vitro* and in xenogeneic mouse models of neuroblastoma (NB).

Results: Co-expression of IL-15 with either costimulatory domain increased CAR-NKT absolute numbers. However, constructs containing 4–1BB induced excessive activation-induced cell death and reduced numeric expansion of NKTs compared with respective CD28-based constructs. Further evaluation of CD28-based GD2.CAR and GD2.CAR.15 showed that co-expression of IL-15 led to reduced expression levels of exhaustion markers in NKTs and increased multi-round *in vitro* tumor cell killing. Following transfer into mice bearing NB xenografts, GD2.CAR.15

Corresponding author: Leonid S. Metelitsa, Department of Pediatrics, Baylor College of Medicine, 1102 Bates Ave, C.1760.06, Houston, TX 77030, lsmeteli@txch.org.

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NKTs demonstrated enhanced *in vivo* persistence, increased localization to tumor sites, and improved tumor control compared to GD2.CAR NKTs. Importantly, GD2.CAR.15 NKTs did not produce significant toxicity as determined by histopathological analysis.

Conclusions: Our results informed selection of the CD28-based GD2.CAR.15 construct for clinical testing and led to initiation of a first-in-human CAR-NKT cell clinical trial ().

Introduction

Recent clinical trials have demonstrated that T cells expressing CD19-specific chimeric antigen receptors (CARs) induce sustained complete responses in patients with B cell malignancies, leading to recent FDA approval of CD19-specific CAR T cell therapies (1-3). However, clinical results obtained using CAR-redirection immunotherapy for solid tumors have been largely disappointing (4,5). Hence, there is an urgent need for alternative strategies that improve the efficacy of CAR-mediated cancer immunotherapy in a wider range of cancers.

CARs can be expressed in T cell subsets with defined functions. For instance, CARs have been expressed in cytotoxic T lymphocytes (CTLs) specific for viral antigens such as those derived from the Epstein Barr Virus (6). Infusion of CTLs expressing a GD2-specific CAR (GD2.CAR) derived from the 14G2a monoclonal antibody to children with neuroblastoma (NB) was proven safe and achieved complete tumor responses in three of 11 patients with refractory/relapsed disease evaluated in one study (7,8). However, CAR-CTLs did not effectively infiltrate tumors or persist *in vivo*, observations that were associated with tumor recurrence or lack of objective responses in the majority of patients. An attempt to increase affinity of GD2.CAR via mutation of 14G2a single chain variable fragment (scFv) enhanced antitumor activity of CAR T cells against NB xenografts, but also produced lethal central nervous system toxicity due to recognition of low GD2-expressing normal brain cells in mice (9). New lines of research are examining CAR-based therapies for NB and other solid tumors using innate and innate-like lymphocytes with natural tumor-trafficking and tumor-targeting properties including NK (10), $\gamma\delta$ T (11), and V α 24-invariant natural killer T (NKT) cells (12).

NKTs are an evolutionarily conserved sub-lineage of innate-like T cells that express the V α 24-J α 18 invariant TCR α -chain and are characterized by reactivity to self- and microbial-derived glycolipids presented by the monomorphic HLA class-I-like molecule CD1d (13). Although the majority of solid tumors are CD1d-negative, the antitumor potential of NKTs has been demonstrated in numerous cancer models (13). We have reported that NKTs actively localize to tumors in response to tumor-derived chemokines CCL2 and CCL20 and that the presence of NKTs within the primary tumor site is associated with improved outcomes in children with NB (14,15). We have also shown that NKTs can be isolated from the peripheral blood, transduced with CARs, and expanded to clinical scale for adoptive cell therapy applications (12,16).

CD19-specific CAR T cells with either CD28 or 4-1BB endodomains have been shown to be effective in patients with B cell malignancies (2). In NB patients, a GD2-specific CAR containing both CD28 and OX40 endodomains enhanced CAR T cell expansion in the

peripheral blood compared to a CAR without costimulation, but the magnitude of expansion was still limited (17). Furthermore, Long *et al.* reported that, unlike CD19-specific CARs, GD2.CARs undergo tonic signaling resulting from spontaneous dimerization caused specifically by the 14G2a scFv. This unintended signaling, which led to early exhaustion of GD2.CAR T cells, was exacerbated in CARs containing the CD28 endodomain and attenuated in CARs containing 4-1BB (18).

We recently described a modified GD2.CAR containing a truncated form of the 14G2a V_H that despite retaining CD28 induces minimal tonic signaling and mediates enhanced antitumor activity in transduced T cells compared to the original GD2.CAR (19). In the same study, we also demonstrated that inclusion of IL-15 in the optimized construct further enhanced the *in vitro* functionality and *in vivo* therapeutic potency of CAR.GD2 T cells in NB models.

Mouse studies and *in vitro* human experimental models have both highlighted the central role of IL-15 in NKT cell development and homeostatic maintenance (20,21). Importantly, IL-15 protects human NKTs from hypoxia, and transgenic expression of IL-15 in adoptively transferred NKTs significantly enhances their antitumor activity (15). Therefore, we hypothesized that co-expressing IL-15 with an optimized GD2.CAR would enhance the survival and antitumor effector functions of NKTs within NB tissues, leading to sustained tumor control. Our results reveal that GD2.CAR constructs encoding 4-1BB, but not CD28, costimulation triggered excessive activation-induced cell death (AICD) in NKTs during *in vitro* expansion. Importantly, co-expression of IL-15 with the GD2.CAR containing the CD28 costimulatory endodomain increased the *in vivo* persistence and antitumor efficacy of CAR-NKTs in metastatic NB models without causing evident toxicity. These preclinical assessments were instrumental in the implementation of first-in-human CAR-NKT clinical testing ().

Materials and Methods

Cell lines and culture conditions

CHLA-255, CHLA-136, LA-N-1, and LA-N-6 cell lines were established and maintained as previously described (15,22,23) and as detailed in Supplemental Methods. 293T cells were obtained from the American Type Culture Collection (ATCC). All cell lines were STR fingerprinted at MD Anderson Cancer Center within one year of use and regularly validated to be free of mycoplasma.

NKT cell isolation, expansion, and transduction

NKT cells were isolated from healthy human donor peripheral blood, *ex vivo* expanded, and transduced with retroviral vectors encoding CAR constructs as previously described (24).

CAR constructs and retroviral vector production

SFG retroviral vectors were constructed as previously described (19) and used to produce retroviral supernatants. Retroviral supernatants were produced by transfecting 293T cells with a combination of an anti-GD2 CAR-containing plasmid, the RDF plasmid encoding the

RD114 envelope, and the PeqPam3 plasmid encoding the MoMLV *gag-pol* as previously described (25). GeneJuice® reagent (Novagen) was used to assist with transfection. Viral supernatants were collected after 48 and 72 hours, filtered with 0.45 µM filters, and frozen.

Flow cytometry

Immunophenotyping was performed using the monoclonal antibodies (mAbs) and reagents detailed in Supplemental Methods. GD2.CARs were detected with the 14G2a anti-idiotypic 1A7 mAb. Analysis was performed on an LSR-II 5-laser flow cytometer (BD Biosciences) using BD FACSDiva software version 6.0 and FlowJo 10.1 (Tree Star, Ashland, OR). SPICE software was used to evaluate expression of exhaustion markers (26).

Cytotoxicity assays

Short-term cytotoxicity of parental and GD2.CAR NKTs against CHLA-255, CHLA-136, LA-N-1, and LA-N-6 cells was evaluated using a four-hour luciferase assay as previously described (16). The calcein-AM M2 macrophage cytotoxicity assay was performed as previously described (27) and detailed in the Supplemental Methods. For the long-term cytotoxicity assay, GFP-tagged CHLA-255, CHLA-136 and LA-N-1 cells were plated in a 24-well plate one day prior to adding NKT cells at a 1:1 effector-to-target ratio. 50 U/mL IL-2 were added every other day. Cells were quantified by flow cytometry at day 7 based on NKT invariant TCR and GFP expression, respectively, with non-transduced NKTs as negative control.

Serial Tumor Challenge Assay

CHLA-255, LA-N-1, CHLA-136 cells (0.5×10^6) and CAR NKT cells (0.5×10^6) transduced with GD2.28z or GD2.28z.15 CAR were co-cultured in a 24-well plate using fresh culture medium with 50 U/mL IL-2, respectively. Five days later, cells were harvested, quantified by trypan-blue exclusion method, and analyzed for NKT and NB cell markers by flow cytometry. CAR NKT cells were then replated at a 1:1 E:T ratio with fresh NB cells in fresh cell culture medium to start the next round tumor co-cultures. At the conclusion of the 4th cycle, NKT cells were counted, and the co-culture was analyzed by flow cytometry.

Multiplex cytokine quantification assay and ELISA

Culture supernatants of NKTs alone or after 48 hours of co-culture with CHLA-255 were collected to measure IL-15 release using the Human IL-15 ELISA Kit (R&D Systems). Supernatants from *in vitro* cultures of NKTs with and without CHLA-255 cells were collected after 24 hours and analyzed for expression of IFNγ, IL-4, GM-CSF, TNFα, and IL-10 using the MILLIPLEX MAP Human Cytokine/Chemokine Immunoassay kit (Millipore) for Luminex® analysis. Mouse IL-6 levels in the plasma were determined using the mouse IL-6 ELISA kit (R&D Systems).

In vivo studies

NSG mice were obtained from The Jackson Laboratory and maintained at the BCM animal care facility. For *in vivo* persistence experiments, mice were injected intravenously (IV) with 1×10^6 CHLA-255 cells. NKTs were co-transduced with a GD2.CAR and firefly luciferase

using retroviral constructs. Seven days after tumor engraftment, approximately 5×10^6 CAR-NKs or control NKs were injected IV into tumor-bearing mice and monitored using bioluminescent imaging every two days (Small Animal Imaging core facility, Texas Children's Hospital). At the time of euthanasia, blood, spleen, liver, lung, and bone marrow were collected and cells were processed for flow cytometry analysis. For *in vivo* therapeutic experiments, mice were injected IV with 1×10^6 firefly luciferase-labeled CHLA-255, CHLA-136, or 2×10^6 LA-N-1 NB cells. On day seven, mice were treated with 5×10^6 CAR-NKs followed by intraperitoneal (IP) injection of IL-2 (2000 U/mouse) every other day for two weeks. Tumor growth was assessed twice weekly by bioluminescent imaging. Animal experiments were performed according to IACUC-approved protocols.

Immunofluorescent microscopy analyses

Livers of non-transduced, IL-15, GD2.28z, and GD2.28z.15 mice were cryosectioned, fixed, and immunostained with anti-human V α 24J α 18 TCR Biotin (clone 6B11, eBioscience), streptavidin Alexa Fluor® 488 conjugate (ThermoFisher), and anti-tyrosine hydroxylase Alexa Fluor® 555 conjugate (clone LNC1, Millipore). Slides were co-stained with ProLong Gold Antifade Reagent with DAPI (ThermoFisher) and images were obtained using an LSM 780 Confocal Microscope (Zeiss).

Statistics

For *in vitro* and *in vivo* experiments, two-sided unpaired or paired Student's t tests were used to evaluate differences in continuous variables between two groups. One-way ANOVA with post-test Bonferroni correction was used for continuous variables among more than two groups, and two-way ANOVA was used when two independent variables were considered. Survival was analyzed using the Kaplan-Meier method and the log-rank (Mantel-Cox) test to compare two groups. Statistics were computed using GraphPad Prism 7.0 (GraphPad Software). Differences were considered significant when the P value was less than 0.05.

Results

Generation and functional testing of NKs expressing CAR constructs with or without IL-15

We generated retroviral constructs incorporating an optimized 14G2a scFv, human CD8 α -derived hinge and transmembrane domains, a human CD28 (GD2.28z) or 4-1BB (GD2.BBz) costimulatory endodomain, and a human CD3 ζ signaling domain ((19) and Supplementary Fig. S1A). The cDNA sequence encoding human IL-15 was added to GD2.28z (GD2.28z.15) or GD2.BBz (GD2.BBz.15) CAR constructs following a self-cleaving viral 2A peptide sequence, allowing production of separate peptide fragments from a single mRNA. Primary human NKs were expanded *ex vivo* following the procedure outlined in Supplementary Fig. S1B and stably transduced with each of the four constructs. CAR expression in NKs was similar for all constructs as assessed three days post-transduction (Fig. 1A). NKs expressing GD2.28z.15 and GD2.BBz.15 secreted IL-15 following a 48-hour culture without or with GD2-positive CHLA-255 NB cells, with higher levels observed in the latter condition following NK activation by CAR engagement (Supplementary Fig. S1C,D).

GD2.28z and GD2.28z.15 NKTs had on average 1.25- and 1.14-fold lower expansion rates than non-transduced (NT) NKTs, respectively at day 10 after secondary stimulation (Fig. 1B). GD2.BBz and GD2.BBz.15 NKT expansion rates dropped to 2.71- and 1.67-fold lower on average than those of NT NKTs, respectively, and were also significantly lower than those of their respective CD28-based counterparts ($p < 0.0001$). When only CAR-positive cells were considered, IL-15 expression increased absolute numbers of NKTs expressing both CD28- and 4-1BB-based constructs (Fig. 1C). Flow cytometry analysis of CAR-NKTs five days post-transduction revealed a significantly higher proportion of early apoptotic cells (annexin-V-positive/7-AAD-negative) in GD2.BBz NKTs versus GD2.28z NKTs ($53.4\% \pm 8.8\%$ vs. $25.9\% \pm 7.45\%$, $p < 0.01$) and in GD2.BBz.15 NKTs vs. GD2.28z.15 NKTs ($28.7\% \pm 8.14\%$ vs. $8.2\% \pm 2.88\%$, $p < 0.05$) (Fig. 1D,E). Furthermore, 4-1BB-CAR-NKTs expressed markedly elevated levels of TNF-related apoptosis-inducing ligand (TRAIL) (Fig. 1F) and FAS (Supplementary Fig. S2A) compared to CD28-CAR-NKTs. We also found that all CAR-NKT groups expressed death receptor TRAIL-R1 (DR4) and to a lesser extent FASL (Supplementary Fig. S2A,B), which together with TRAIL and FAS respectively could lead to self-initiated death receptor signaling in these cells. Indeed, we found higher levels of death receptor-dependent caspase-8 activity in 4-1BB-CAR-NKTs than CD28-CAR-NKTs (Fig. 1G). Thus, the 4-1BB costimulatory endodomain induces AICD, leading to loss of CAR-NKTs during *ex vivo* expansion. Instead, NKTs transduced with CD28-containing GD2.CAR constructs—particularly with IL-15—undergo relatively low levels of AICD and are capable of extensive numeric expansion, making them a superior option for generating clinical-scale CAR-NKT cell products. Therefore, we selected the GD2.28z and GD2.28z.15 constructs for further pre-clinical evaluation.

Co-expression of IL-15 enhances survival and reduces exhaustion-associated phenotype in GD2.CAR NKTs

Co-expression of IL-15 with a CAR construct should provide a survival advantage for effector cells in the immunosuppressive tumor microenvironment. To determine whether transgenic IL-15 co-expressed with the GD2.CAR mediates this intended effect, we cultured GD2.28z and GD2.28z.15 NKTs in cytokine-deprived serum-free medium and evaluated key molecules of the IL-15-mediated pro-survival signaling pathway. We found that GD2.28z.15, but not GD2.28z NKTs robustly phosphorylated Stat5 (Fig. 2A) and upregulated expression of Bcl-2 family members Bcl-2 (Fig. 2B) and Bcl-xL (Fig. 2C), indicating that transgenic IL-15 recapitulates the pro-survival functionality of the natural cytokine.

Next, we examined expression trends for major human NKT functional markers CD4 and CD62L in GD2.28z and GD2.28z.15 NKTs at the end of *in vitro* expansion. We confirmed previously reported inter-individual variability in CD4 expression across all donors tested (28). NKTs from donors with high frequency of CD4⁺ cells (80 – 100%) retained high proportion of CD4⁺ NKTs after CAR transduction (Fig. 2D). In contrast, NKTs from donors with low to intermediate levels of CD4 expression (20 – 50%) either decreased or increased the proportion of CD4⁺ cells after CAR transduction. We detected a consistent decrease in the proportion of CD62L⁺ NKTs in CAR-transduced versus NT cells, which is likely due to a residual tonic signaling induced by GD2.CAR, a common phenomenon in the currently used CARs (29). However, GD2.28z.15 NKTs retained a significantly higher proportion of

CD62L⁺ cells compared to GD2.28z NKTs in 9 of 11 donors (Fig. 2E). Evaluation of exhaustion marker levels revealed that CAR expression led to significant upregulation of PD-1, TIM-3, and LAG-3 in NKTs (Fig. 2F). However, GD2.28z.15 NKTs expressed significantly lower levels of PD-1, TIM-3, LAG-3, and TIGIT (Fig. 2F) and were characterized by significantly fewer cells positive for three of the exhaustion-associated markers than GD2.28z NKTs (Fig. 2G). There was no significant difference in the level of CAR expression observed between the GD2.28z and GD2.28z.15 NKTs ($p > 0.05$, paired t-test, $n=6$). The addition of soluble IL-15 to NKT cell culture medium for 7 days after TCR stimulation resulted in a similar PD-1 downregulation in non-transduced and GD2.28z NKTs (Supplementary Fig. S3E). Therefore, co-expression of IL-15 with the GD2.CAR enhances NKT survival and alleviates the exhaustion phenotype.

Co-expression of IL-15 enhances *in vitro* functional fitness of GD2.CAR NKTs

To determine the efficacy of CAR-mediated killing by NKT cells, we co-cultured GD2.CAR NKTs with firefly luciferase (Ffluc)-expressing CD1d-negative and GD2-high (CHLA-255 and LA-N-1), GD2-medium (CHLA-136), or GD2-negative (LA-N-6) NB cells (Fig. 3A) for four hours. NKTs expressing either GD2.28z or GD2.28z.15 did not kill GD2-negative LA-N-6 cells, but mediated equally potent short-term cytotoxicity against GD2-high NB cell lines (Fig. 3B). GD2-medium CHLA-136 cells were killed less effectively than GD2-high lines in four hours, but were eliminated by either GD2.28z or GD2.28z.15 NKTs within 24 hours (Fig. 3C). While parental NKTs did not kill CD1d-negative NB cells, GD2.CAR NKTs retained their native CD1d-restricted reactivity, as they killed α -galactosylceramide-pulsed M2-polarized macrophages (GD2-negative/CD1d-positive) as effectively as parental NKTs (Fig. 3D). Next, we examined changes in GD2.CAR NKT numbers and phenotype after challenge with NB cells. Seven days after adding any of three NB cell lines, GD2.28z.15 mediated expansion/survival of significantly higher numbers of NKTs compared with GD2.28z. However, challenge with CHLA-136 cells resulted in fewer GD2.CAR NKTs compared to CHLA-255 or LA-N-1 cells (Fig. 3E).

To directly test functional fitness, we co-cultured GD2.CAR NKTs with GFP-transduced NB cells for five days, then added equal numbers of the effector cells to fresh tumor cells for another five days and repeated such settings in four rounds. At the end of each round, viable NKT and NB cells were analyzed by flow cytometry (Supplementary Fig. S3A,B). We found that GD2.28z.15 mediated greater NKT cell numeric expansion after each round compared with GD2.28z (Fig. 3F). While NKTs expressing either CAR construct effectively eliminated NB cells in the first three rounds, GD2.28z.15 was more effective in controlling tumor cell numbers compared with GD2.28z at the end of the fourth round (Fig. 3G). The flow cytometry analysis of NKTs after the fourth round revealed higher levels of PD-1 and TIGIT expression in NKTs transduced with GD2.28z compared to GD2.28z.15 (Supplementary Fig. S3C), suggesting that the presence of IL-15 in CAR construct attenuates NKT-cell exhaustion that may enhance their functionality. Similar results were obtained in a repeat challenge experiment using an IncuCyte instrument which allows continuous measurement of viable tumor cells in co-culture with CAR NKTs (Supplementary Fig. S3D).

To further evaluate the functional potential of GD2.CAR NKTs, we determined their cytokine production profiles in response to stimulation with CHLA-255 NB cells. GD2.28z and GD2.28z.15 NKTs both produced high levels of T_H1 cytokine interferon- γ (IFN γ) in the presence CHLA-255 cells, with levels approximately two-fold higher in GD2.28z.15 NKTs (Fig. 3H). Both groups produced similar levels of T_H2 cytokine IL-4, resulting in an overall higher IFN γ -to-IL-4 ratio in GD2.CAR NKTs with IL-15 (Fig. 3I). Similar trends were observed with T_H1 cytokines GM-CSF and TNF α , while IL-10 was not detected from either GD2.CAR NKT group (Supplementary Fig. S4).

GD2.28z.15 construct supports superior NKT cell *in vivo* persistence

Long-term persistence of CAR-effector cells is critical to the success of adoptive cell therapies (30-33). We examined the *in vivo* persistence of GD2.28z and GD2.28z.15 NKTs co-transduced with high-intensity Ffluc cDNA following transfer into NSG mice with 7-day established NB xenografts (Fig. 4A). Mice injected with Ffluc-only and IL-15/Ffluc NKTs served as control groups. Transgenic expression of IL-15 on its own improved NKT persistence compared to NT/Ffluc NKTs while GD2.28z only marginally boosted persistence above NT/Ffluc control (Fig. 4B,C). Importantly, addition of IL-15 to the GD2.28z construct significantly enhanced the ability of infused NKTs to expand and persist beyond levels seen in GD2.28z or IL-15 NKTs. We checked the bone marrow, the primary site of NB metastases in humans and our mouse model, for the presence of tumor and NKTs 20 days after NKT cell transfer. Both IL-15/Ffluc and GD2.28z.15 NKT cells were present at approximately six times higher frequency in the bone marrow than NT/Ffluc NKTs, while GD2.28z NKT frequency did not exceed that of control NKTs (Fig. 4D). Accordingly, mice treated with GD2.28z.15 NKTs had significantly fewer NB cells in the bone marrow than mice treated with GD2.28z NKTs (Fig. 4E).

GD2.28z.15 construct mediates effective infiltration of NKTs into solid tumor tissues

To control tumor growth, therapeutic effector cells must effectively traffic to the tumor site. Using tumor tissues from NB nodules in the livers of NSG mice (Fig. 5A), we performed confocal microscopy to evaluate the ability of GD2.CAR NKTs to infiltrate the tumor 13 days after treatment. Consistent with previous observations (12,15), we found that transgenic expression of either IL-15 alone or a GD2.CAR significantly increased the frequency of NKTs in the tumor tissue as compared to control NKTs (Fig. 5B,C). GD2.28z.15 NKTs were present at significantly higher numbers within NB nodules than both GD2.28z NKTs and IL-15-only NKTs (Fig. 5B,C). In addition to the tumor, GD2.28z and GD2.28z.15 CAR-NKTs were detected in the spleen, liver, bone marrow, and lung (Fig. 5D). Within these tissues, GD2.28z.15 NKTs were predominantly CD4⁻ compared to GD2.28z NKTs (Fig. 5E), indicating that local IL-15 production preferentially benefits survival and/or expansion of CD4⁻ over CD4⁺ NKTs. CD62L expression was generally low in GD2.CAR NKTs infiltrating the tissues regardless of transgenic IL-15 (Fig. 5F). At the later time point (8 weeks) after infusion, CAR-NKTs could still be found in bone marrow at the frequencies of 6.26% $\pm$ 1.006% and 0.026% $\pm$ 0.003% for GD2.28z.15 and GD2.28z, respectively (Fig. 5G, $p < 0.001$). Again, these long-term persisting GD2.28z.15 NKTs were preferentially CD4⁻ cells in three experiments from NKT-cell donors with 20 – 50% CD4⁺ NKT frequency in peripheral blood (Fig. 5H). However, GD2.28z.15 NKTs from one donor with high

proportion of CD4⁺ subset at baseline still exhibited increased persistence in bone marrow while retaining CD4 expression (Supplementary Fig. S5A,B). Therefore, GD2.28z.15 NKTs effectively home to and persist in tumor tissues, in particular due to increased survival and/or expansion of the CD4⁺ subset with a possible exception for donors with high proportion of CD4⁺ NKTs.

GD2.28z.15 construct enables superior therapeutic activity of GD2.CAR NKTs in mice without significant toxicity

We next evaluated the ability of GD2.CAR NKTs to control tumor growth in three xenogeneic NB models. NSG mice were intravenously injected with CHLA-255, LA-N-1, or CHLA-136 NB cells stably expressing Ffluc, and seven days later they received 5×10^6 of either GD2.28z or GD2.28z.15 NKTs intravenously (Fig. 6A). Mice treated with NT or IL-15-transduced NKTs were used as control groups. We found that GD2.28z.15 NKTs controlled tumor growth significantly longer than GD2.28z NKTs with all three cell lines, ($p < 0.001$) (Figs. 6B-J). The GD2.28z.15 NKT cell therapy was particularly effective against xenografts of two NB cell lines with high level GD2 expression, achieving a prolonged survival without evidence of disease in the majority of mice at the time of sacrifice at 3 months. We also treated CHLA-255 tumor-bearing mice with a lower (2×10^6) or a higher (10^7) dose of CAR NKTs. Supplementary Fig. S6 demonstrates that NKTs expressing IL-15-containing CAR retain superior antitumor activity at a low and a high dose. Moreover, only GD2.28z.15 NKT therapy achieved a long-term disease free survival in the majority of mice at the high dose. Thus, consistent with its ability to mediate the longest duration of *in vivo* persistence and most effective tumor infiltration, GD2.28z.15 also imparted NKTs with superior dose-dependent antitumor activity, especially against tumors with high level GD2 expression.

We observed mild hunched mouse stature and ruffled fur twelve days post-injection of GD2.28z.15 NKTs that completely resolved within two-to-three days. At approximately the same time, transient weight loss not exceeding 10% of total body weight also occurred (Supplementary Fig. S7A). Beyond day 14, animals maintained or increased weight. Serum collected on days 12 and 15 was analyzed by ELISA for the presence of human IL-15 and murine IL-6, the latter of which is produced predominantly by murine macrophages in a manner that parallels development of cytokine release syndrome in humans (34). Murine IL-6 levels were not significantly different between GD2.28z.15 NKT-treated and untreated control mice (Supplementary Fig. S7B), while human IL-15 remained undetectable in both groups. Detailed pathological examination was also performed by an experienced clinical pathologist in mice sacrificed at day 12 or 15 post-NKT injection. Lung, heart, liver, spleen, kidney, adrenals, stomach, intestine, pancreas, spine, brain (cerebrum, hippocampus, and cerebellum), skin, bone with bone marrow, and skeletal muscle revealed minimally detectable tumor cells in two mice. Additionally, no pathologic abnormalities were detected in any organ or tissue with the exception of the lungs, where mice in both groups exhibited signs of focal minimal-to-mild interstitial inflammation. An additional long-term tissue evaluation performed between 45 and 60 days post-injection of GD2.28z.15 NKTs also showed complete absence of toxicity or features of graft-versus-mouse disease (Supplementary Fig. S7C). There was also no tissue toxicity observed including that in brain

in mice treated with 10^7 GD2.28z.15 NKTs (Supplementary Fig. S8). Thus, GD2.28z.15 NKTs exert potent anti-NB activity without causing significant off-target toxicity in mice.

Discussion

NKTs are a promising cellular platform for CAR-redirection cancer immunotherapy. Our previous studies have demonstrated that NKTs naturally traffic to NB in xenograft models in response to tumor-derived chemokines and that their ability to infiltrate tumors is further enhanced by expression of a tumor-specific GD2.CAR (12,15).

We have also found that NKT-cell functionality is inhibited in the hypoxic tumor microenvironment and at least in part rescued by the transgenic expression of IL-15 (15). Therefore, we sought to test the hypothesis that co-expression of IL-15 with the GD2.CAR would enhance the ability of NKTs to traffic to and attack NB tumors. Our results demonstrate that NKTs expressing either GD2.28z or GD2.28z.15, but not GD2.BBz or GD2.BBz.15 CAR constructs can be expanded *ex vivo* to clinical scale and mediate potent CAR-dependent killing of NB cells *in vitro*. Compared to GD2.28z, GD2.28z.15 CAR NKTs demonstrated superior functional fitness, prolonged *in vivo* persistence, and superior tumor control in mice without causing significant toxicity.

When comparing GD2.CAR constructs incorporating the two most commonly used costimulatory endodomains, CD28 and 4-1BB, we found that 4-1BB-containing constructs induced higher levels of AICD in NKTs than CD28-containing constructs, regardless of IL-15 co-expression. This AICD was associated with increased TRAIL and FAS expression, activation of caspase-8, and induction of apoptosis, resulting in lower overall NKT cell numeric expansion. In line with our findings in NKTs, Mamonkin *et al.* observed that 4-1BB-containing CARs expressed from a retroviral vector with strong LTR promoter activity enhanced AICD levels in transduced T cells (35). Specifically, 4-1BB-mediated TRAF-dependent NF κ B activation led to upregulation of death receptors and an increased rate of AICD in CAR T cells.

Although GD2.28z and GD2.28z.15 NKTs were equally effective in short-term cytotoxicity assays against two GD2-high (CHLA-255 and LA-N-1) NB cell lines, both mediated reduced cytotoxicity against GD2-medium (CHLA-136) NB cells. CHLA-255 is a MYCN-non-amplified line while LA-N-1 and CHLA-136 are MYCN-amplified lines, suggesting that the efficacy of CAR.GD2 NKT cell therapy depends on the level of GD2 expression on NB cells rather than their MYCN status. GD2.28z.15 NKTs demonstrated significantly more effective tumor control in a repeat challenge assay. This activity was associated with expansion of GD2.28z.15 but not GD2.28z NKTs. We recently showed parallel results with T cells expressing the same two constructs; GD2.28z.15 T cells demonstrated superior expansion and antitumor activity after numerous tumor cell exposures compared to GD2.28z T cells (19). Additionally, we have found that expression of IL-15-containing CARs is associated with reduced exhaustion marker expression in both T and NKT cells, suggesting that IL-15 prevents NKT-cell exhaustion in the same manner as it does in T cells.

Co-expression of IL-15 with the GD2.CAR enriched for T cells with memory and stem-cell-like phenotypes (19). While the mechanisms governing memory differentiation of human NKTs remain less explored than those of T cells, we recently demonstrated that CD62L⁺ central memory-like NKTs can expand in culture and are largely responsible for the *in vivo* persistence and therapeutic potential of CAR-NKTs (16). The addition of IL-15 to the CAR increased the frequency of CD62L⁺ NKTs in 9 of 11 donors examined, suggesting that IL-15 protects NKTs from cell death and exhaustion, and may provide a selective advantage for central memory-like NKTs.

After adoptive transfer to tumor-bearing mice, GD2.28z.15 NKTs demonstrated remarkable persistence especially in the bone marrow and liver (primary sites of NB metastasis) despite the highly hypoxic microenvironment of NB liver metastases that hampers NKT survival and function (15). We previously reported that transgenic IL-15 protects NKTs from hypoxia (15), and here we further observed that IL-15 promotes survival of CAR-NKTs in NB xenografts in the liver. Furthermore, phenotypic analysis of infiltrated CAR-NKTs revealed enrichment of NKTs expressing low levels of CD62L, suggesting that persistent CAR stimulation promotes effector-memory-like differentiation of NKTs as occurs in T cells.

NKTs produce numerous cytokines, which individually may exert either pro-inflammatory or inhibitory effects. Our *in vitro* testing showed that NKTs co-expressing IL-15 and a GD2.CAR predominantly secrete a T_H1-like profile following CAR engagement as evidenced by increased production of IFN- γ , GM-CSF, and TNF α as well as an increased IFN- γ to IL-4 ratio. This observation is consistent with our previous findings showing higher expression of the IL-2R β /IL-15R β subunit (CD122) in CD4⁻ NKTs (21), which produce more T_H1 and less T_H2 cytokines than CD4⁺ NKTs (28). Moreover, in murine models, CD4⁻ NKTs have been reported to have better antitumor functionality *in vitro* and *in vivo* compared to their CD4⁺ counterparts (36). Despite increased sensitivity of CD4⁻ NKTs to IL-15, we did not observe consistent changes in CD4⁺/CD4⁻ proportion during *in vitro* expansion of GD2.28z and GD2.28z.15 NKTs, likely due to the presence of IL-2 in culture medium. In contrast, after adoptive transfer to tumor-bearing mice, GD2.28z.15 NKTs derived from 3 donors with low to intermediate levels of CD4⁺ cells persisted predominantly as CD4⁻ cells within the tumor. We did observe, however, that GD2.28z.15 NKTs derived from a donor with high proportion of CD4⁺ NKTs retained CD4 expression *in vivo*, a phenomenon that needs to be further explored.

We found that the potent antitumor effect of GD2.28z.15 NKTs occurred without significant toxicity. The pathological analysis of mouse tissues including brain did not reveal signs of inflammation. In contrast, Richman et al. observed lethal CNS toxicity when testing T cells expressing a mutated CAR.GD2 with an increased affinity to GD2 (9), indicating that increasing CAR affinity to GD2 may not be a viable strategy for increasing therapeutic efficacy of GD2-targeting immunotherapy. Instead, our approach is based on the enhancing functionality of the CAR-expressing effector cells that leads to enhanced antitumor activity without evident toxicity to normal tissues. In contrast, systemic administration of IL-15 in patients with cancer has been associated with dose-limiting toxicities (37), although IL-15/IL15Ra superagonists such as ALT-803 show an increased therapeutic index in the initial clinical testing (38). IL-15 remained undetectable in the serum of mice treated with

GD2.28z.15 NKTs, indicating that the therapeutic effects of IL-15 are likely associated with local release of the cytokine at the tumor site.

Based on results of a randomized phase III clinical trial (39), FDA approved anti-GD2 monoclonal antibody ch14.18 (dinutuximab) in combination with GM-CSF, IL-2, and 13-cis-retinoic acid for maintenance therapy of high-risk NB. The antitumor activity of dinutuximab largely depends on ADCC, which is mediated by NK and myeloid cells and augmented by IL-2 and GM-CSF, respectively (40,41). Ch14.18 given as a single agent is ineffective (42). While the role of individual cytokines in this combination therapy is still under investigation, the current regimen remains toxic and not curative for a third of the high-risk patients (43). GD2.28z.15 NKT cell therapy is designed to maximize direct neuroblastoma cell killing without dependence on ADCC or systemic cytokine administration that may lead to a more effective and safe immunotherapy of high-risk and relapsed NB.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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Abbreviations list:

AICD	activation-induced cell death
CAR	Chimeric antigen receptor
DR4	death receptor TRAIL-R1
Ffluc	firefly luciferase
GD2.CAR	anti-GD2 CAR
GD2.CAR.15	anti-GD2 CAR with IL-15
GD2.28z	anti-GD2 CAR with CD28 co-stimulatory endodomain
GD2.BBz	anti-GD2 CAR with 4-1BB co-stimulatory endodomain
GD2.28z.15	GD2.28z with IL-15
GD2.BBz.15	GD2.BBz with IL-15
IL-15	interleukin 15
IP	intraperitoneal

IV	intravenous
LAG-3	Lymphocyte-activation gene 3
mAbs	monoclonal antibodies
NB	neuroblastoma
NKT	V α 24-invariant natural killer T
NSG	NOD/SCID/IL-2R γ null
PD-1	Programmed cell death protein 1
scFv	single chain variable fragment
TIGIT	T-cell immunoreceptor with Ig and ITIM domains
TIM-3	T-cell immunoglobulin and mucin-domain containing-3
TRAIL	TNF-related apoptosis-inducing ligand

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Translational Relevance

CAR-redirected T cell adoptive immunotherapy remains largely ineffective in solid tumors due in part to the limited ability of conventional T cells to localize to and survive in tumor tissues. In this work, we exploit the intrinsic tumor-trafficking properties of natural killer T (NKT) cells to home to neuroblastoma tumors and target them with an engineered GD2-specific CAR co-expressing human IL-15 to enhance CAR-NKT survival. We expressed a set of optimized GD2-based CAR constructs with either the human CD28 or 4-1BB costimulatory endodomain—each with or without IL-15—in primary human NKTs and compared the therapeutic potential and toxicity of resultant CAR-NKTs. The results of this study informed the design of an ongoing first-in-human clinical evaluation of CAR-NKT cell therapy in children with neuroblastoma ().

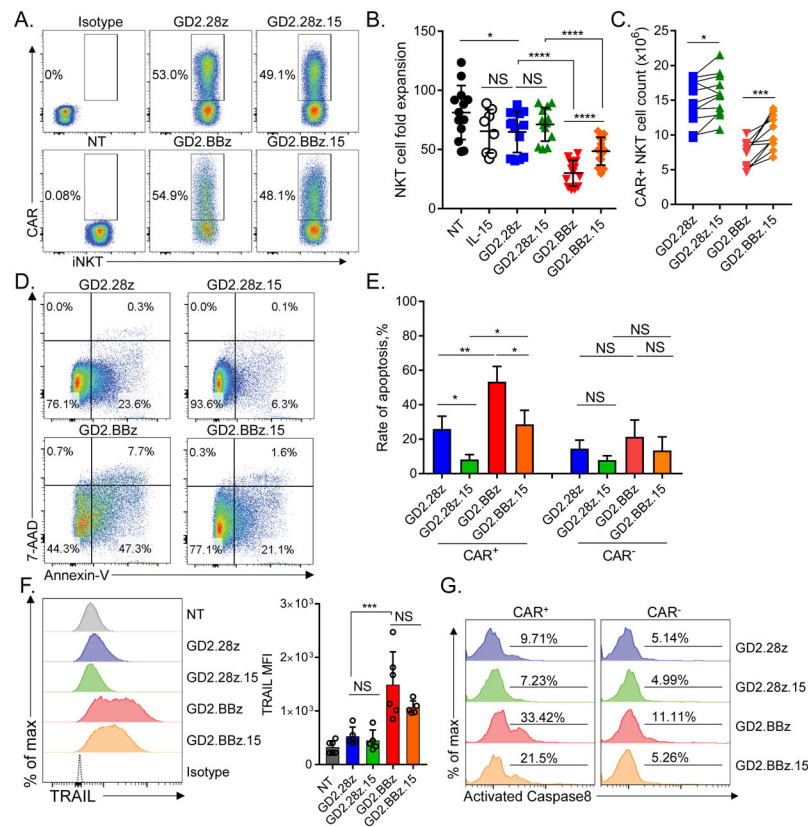


Figure 1. GD2.CAR constructs containing CD28 costimulatory endodomain enable superior ex vivo expansion of CAR-NKs.

(A) Following a 10-day primary stimulation with α -GalCer-pulsed autologous PBMCs, NKs were re-stimulated and transduced 2 days later with the indicated CAR constructs or no construct (NT: non-transduced). Representative flow cytometry analysis of indicated GD2.CAR construct expression 3 days after transduction by gating on 1A7+ (14G2a anti-idiotypic)/ V α 24-J α 18+ NK cells (n=13). (B) Quantification of total NK cell fold expansion, mean $\pm$ SD following 10 days of secondary expansion (n=13). NK cell number was determined using the Cellometer Auto Cell Viability Counter with AOPI staining. *p < 0.05, ****p < 0.0001, NS: not significant, one-way ANOVA. (C) Quantification of CAR+ NK cell expansion calculated following 10 days of secondary expansion. *p < 0.05, *** p < 0.001, Student's paired t-test. (D) NKs transduced with the indicated GD2.CAR constructs were stained with annexin-V and 7-AAD 5 days after transduction. Shown are representative results from one of four donors. (E) Quantification of annexin-V +/7-AAD-NKs $\pm$ SD in both CAR+ gated and CAR- gated populations for indicated GD2.CAR NK groups (n=4). *p < 0.05, **p < 0.01, NS: not significant, one-way ANOVA. (F) TRAIL expression in CAR+ gated populations for the indicated GD2.CAR NKs 7 days post-transduction. Shown are TRAIL median fluorescence intensity (MFI) results from a representative donor and mean $\pm$ SD of TRAIL expression MFI from six donors. ***p < 0.001, NS: not significant, one-way ANOVA. (G) Levels of activated caspase 8 were evaluated in the indicated CAR+ and CAR- NK populations using FAM-LETD-FMK labeling 7 days post-transduction. Shown are representative results from one of three donors.

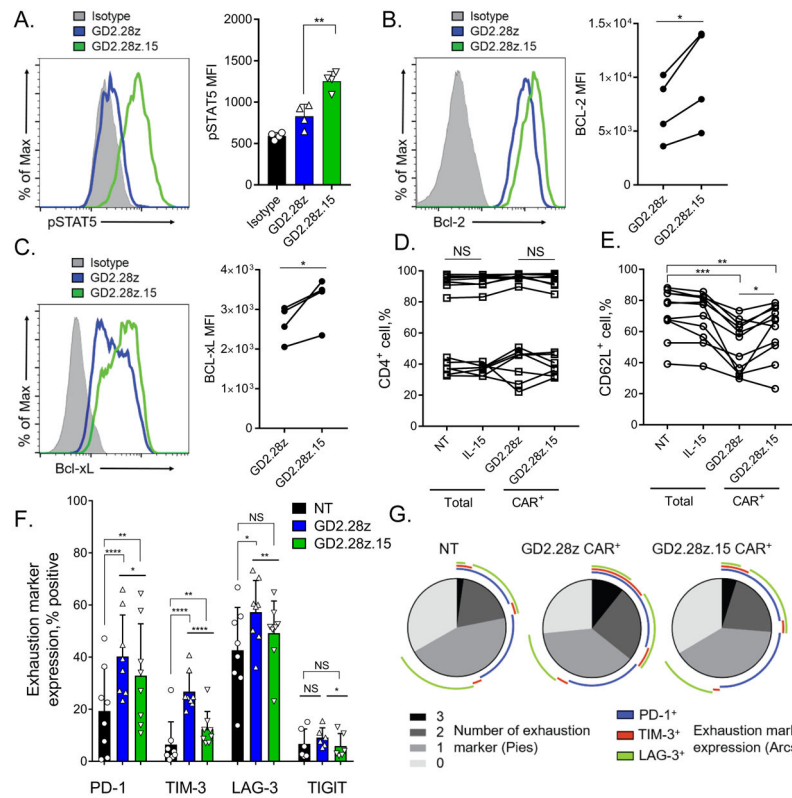


Figure 2. Co-expression of IL-15 enhances survival and reduces exhaustion-associated phenotype in GD2.CAR NKTs.

(A) Expression of phosphorylated STAT5 (pSTAT5) was evaluated in NKTs expressing the GD2.28z or GD2.28z.15 CAR 7 days post-transduction following 12 hours of culture in serum- and cytokine-starved conditions. Shown is a representative histogram from one of four donors (left) and mean $\pm$ SD of MFI for all donors (n=4). **p < 0.01, one-way ANOVA. (B) Expression of intracellular Bcl-2 was evaluated in NKTs expressing the GD2.28z or GD2.28z.15 CAR 7 days post-transduction following 12 hours of culture in serum- and cytokine-starved conditions. Shown is a representative histogram from one of four donors (left) and Bcl-2 expression MFI for each individual donor expressing the GD2.28z versus the GD2.28z.15 CAR (n=4). *p < 0.05, Student's paired t-test. (C) Expression of intracellular Bcl-xL was evaluated in NKTs expressing the GD2.28z or GD2.28z.15 CAR 7 days post-transduction following 12 hours of culture in serum- and cytokine-starved conditions. Shown is a representative histogram from one of four donors (left) and Bcl-xL expression MFI for each individual donor expressing the GD2.28z versus the GD2.28z.15 CAR (n=4). *p < 0.05, Student's paired t-test. (D) Following 21-day *in vitro* expansion, NKTs from 14 individual donors transduced with the indicated constructs or non-transduced (NT) NKTs were evaluated for CD4 expression and (E) 11 of them were also examined for CD62L expression by flow cytometry. NS: not significant, *p < 0.05, **p < 0.01, ***p < 0.001, Student's paired t-test. (F) NKTs transduced with the GD2.28z or GD2.28z.15 CAR or NT were evaluated for expression of the indicated exhaustion markers 7 days after transduction (n=8). *p < 0.05, **p < 0.01, ****p < 0.0001, NS: not significant, one-way ANOVA. (G) SPICE analysis of exhaustion marker expression from three of the

donors shown in F. Pie charts reflect proportions of indicated NKT groups expressing indicated numbers (0-3) of exhaustion markers. Colored arcs indicate the specific combinations of exhaustion markers expressed.

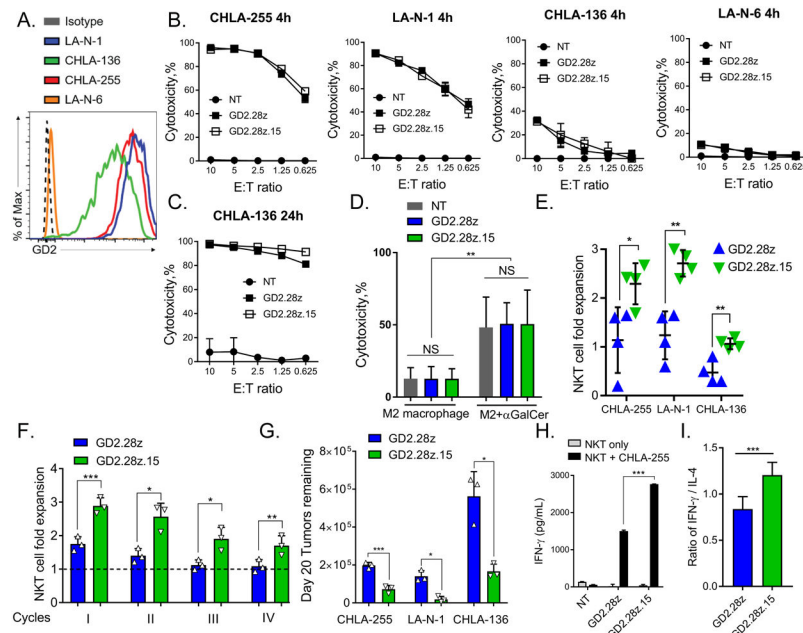


Figure 3. Co-expression of IL-15 enhances the *in vitro* functional fitness of GD2.CAR NKTs. (A) Flow cytometry analysis of GD2 expression in the indicated NB cell lines. (B) The indicated luciferase-transduced NB cell lines were co-cultured with GD2.CAR NKTs for 4 or (C) 24 hours. Cytotoxicity was calculated by measuring luminescence intensity of lysed target cells with a plate reader. Shown are results from a representative of four experiments (each with a separate NKT donor) with three technical replicates. (D) GD2.CAR NKTs were co-cultured with calcein-AM-labeled M2-polarized macrophages with or without α -GalCer for 4 hours. Cytotoxicity was assessed by flow cytometry. Shown are mean $\pm$ SD from four donors. ** $p < 0.01$, NS: not significant, two-way ANOVA. (E) NKT cell fold expansion measured after 1:1 co-culture with NB cells for 7 days with 50 U/mL IL-2. NKT cell number was determined using the cell counter and flow cytometry. Shown are mean $\pm$ SD from four donors. * $p < 0.05$, ** $p < 0.01$, Student's unpaired t-test. (F) NB cells expressing GFP were plated at a 1:1 effector-to-target (E:T) ratio with GD2.CAR NKTs every 5 days for four cycles. NKT cells were harvested from the previous co-culture and then re-plated in new culture medium with fresh tumor cells at the same 1:1 E:T ratio. At each cycle, cells were counted and analyzed by flow cytometry. Shown are results of a representative cell line, CHLA-255, from one of two independent experiments performed with three NKT donors per experiment. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, Student's paired t-test. (G) Tumor cells remaining after cycle 4 following the experiment described in F. * $p < 0.05$, *** $p < 0.001$, Student's paired t-test. (H) GD2.CAR NKTs were stimulated with CHLA-255 cells or left unstimulated and supernatants were collected at 24 hours. IFN- γ concentrations were measured by Luminex assay. Shown are mean $\pm$ SD from a representative of three experiments with three technical replicates. *** $p < 0.001$, two-way ANOVA. (I) Ratio of IFN- γ -to-IL-4 production by GD2.CAR NKTs following 24 hour co-culture with CHLA-255 cells. Shown are mean $\pm$ SD for three donors with three replicates each. *** $p < 0.001$, Student's unpaired t-test.

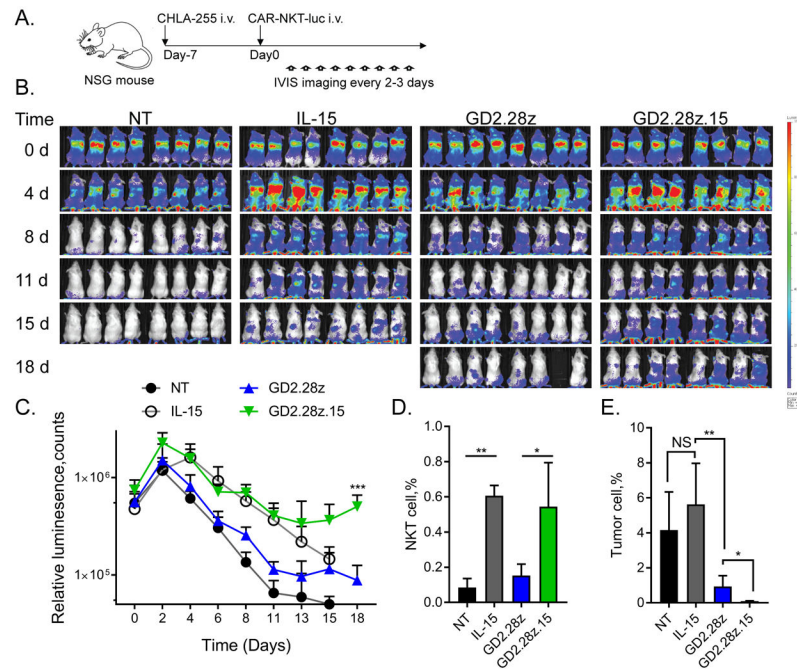


Figure 4. The GD2.28z.15 construct supports superior NKT cell *in vivo* persistence. (A) Overview of *in vivo* NKT persistence experiments. NSG mice were injected intravenously with CHLA-255 neuroblastoma cells followed 7 days later by luciferase-labeled CAR.GD2 or control NKTs. NKTs were tracked by bioluminescence imaging every two-to-three days. (B) Bioluminescent monitoring of NT, IL-15, GD2.28z, and GD2.28z.15 NKTs injected into groups of mice (n=8/group) carrying CHLA-255 xenografts. Shown are representative results from one of three experiments (each with a separate NKT donor). (C) Quantification of bioluminescence images in B. ***p < 0.001, GD2.28z versus GD2.28z.15, Student's unpaired t-test. (D) Quantification of NKT cells (human CD45+/NKT+) in total bone marrow cells at day 20 as determined by flow cytometry. *p < 0.05, **p < 0.01, one-way ANOVA. (E) Quantification of neuroblastoma tumor cells (human CD56+/GD2+) in total bone marrow cells at day 20 as determined by flow cytometry. Five mice in both NT and IL-15 groups were granted exceptions to delay euthanasia until day 20, but could not be imaged on day 18. *p < 0.05, **p < 0.01, NS: not significant, Student's unpaired t-test.

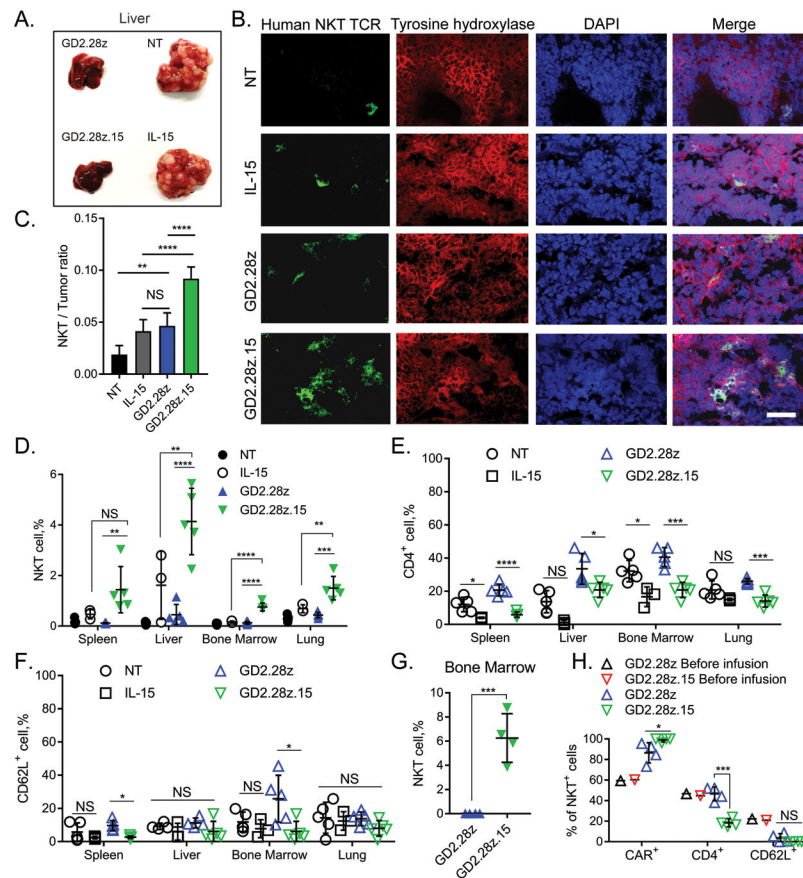


Figure 5. GD2.28z.15 mediates effective infiltration of NKTs into solid tumor tissues and expansion of CD4-negative NKTs. (A) CHLA-255-bearing NSG mice were injected with luciferase-labeled GD2.CAR or control NKTs in a similar experimental setup as shown in Figure 4A. Thirteen days after injection of NKTs, resected livers were photographed. Shown are representative images from 5 mice per group, one of three experiments. (B) Liver metastases from “A” were fixed and processed for cryosectioning. Sections were probed with antibodies against the V α 24-J α 18 NKT TCR, neuroblastoma tumor marker tyrosine hydroxylase, and DAPI. Images were visualized and quantified by confocal microscopy. (C) Absolute numbers of NKT cells and tumor cells were counted in 10 fields per mouse using images in “B”, 5 mice per group. ** $p < 0.01$, **** $p < 0.0001$, NS: not significant, one-way ANOVA. (D) Quantification of human NKT cells (human CD45⁺/NKT⁺) in total cell populations collected from indicated organs (n=5), one of three experiments. ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, NS: not significant, one-way ANOVA. (E) Quantification of CD4⁺ NKT cells in total cell populations collected from indicated organs (n=5). * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$, NS: not significant, one-way ANOVA. (F) Quantification of CD62L⁺ NKT cells in total cell populations collected from indicated organs (n=5). * $p < 0.05$, NS: not significant, one-way ANOVA. (G) At week 8 after injection, human NKT cells were quantified in total cell populations collected from bone marrow (n=4). **** $p < 0.001$, Student’s unpaired t-test. (H) Quantification of CAR⁺, CD4⁺ or CD62L⁺ subsets of NKT cells in bone marrow as in G

(n=4). Data from a representative of three independent experiments. * $p < 0.05$, *** $p < 0.001$, NS: not significant, Student's unpaired t-test.

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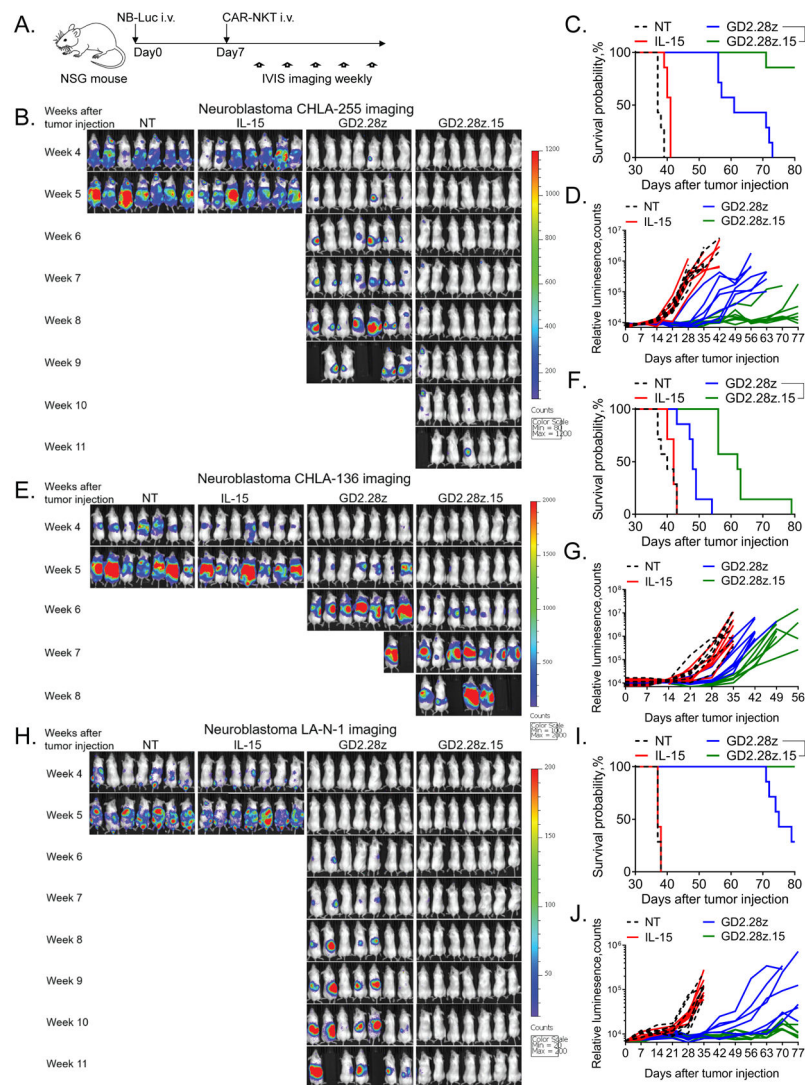


Figure 6. GD2.28z.15 enables superior *in vivo* therapeutic activity of GD2.CAR NKTs without causing significant toxicity.

(A) Schematic representation of *in vivo* tumor challenge experiments. NSG mice were injected intravenously with luciferase-labeled NB cell lines followed 7 days later by CAR.GD2 NKTs. Tumor growth was assessed weekly by bioluminescence imaging. (B) Weekly bioluminescent monitoring of CHLA-255 tumor growth in mice (n=7/group). (C) Kaplan-Meier curve generated from survival of animals in B. ***p < 0.001 Log-rank (Mantel-Cox). (D) Quantification of bioluminescence in B. (E) Weekly bioluminescent monitoring of CHLA-136 tumor growth. (F) Kaplan-Meier curve generated from survival of animals in E. ***p < 0.001 Log-rank (Mantel-Cox). (G) Quantification of bioluminescence in E. (H) Weekly bioluminescent monitoring of LA-N-1 tumor growth. (I) Kaplan-Meier curve generated from survival of animals in H. **p < 0.01 Log-rank (Mantel-Cox). (J) Quantification of bioluminescence in H. Shown are representative results from one of four independent experiments with the CHLA-255 model and one experiment each for the CHLA-136 and LA-N-1 models.